LAPORAN PRAKTIKUM

POSTTEST (6)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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KELAS (C'1)

PROGRAM STUDI INFORMATIKA

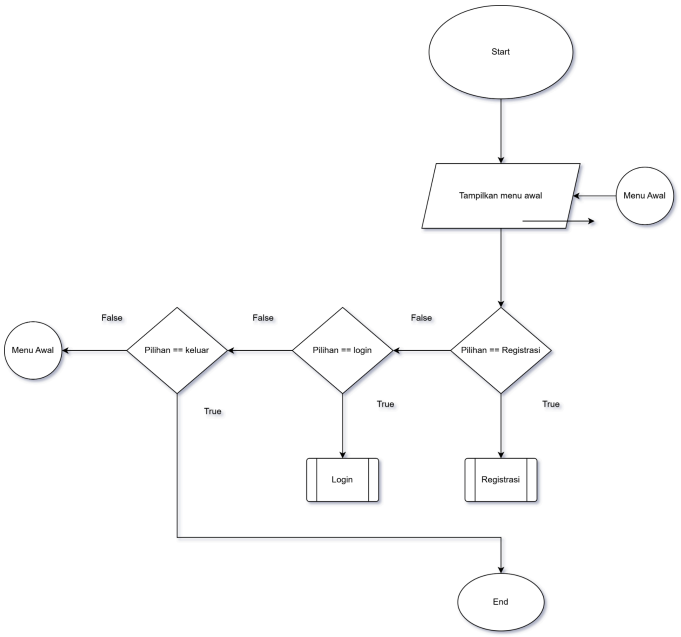
UNIVERSITAS MULAWARMAN

SAMARINDA

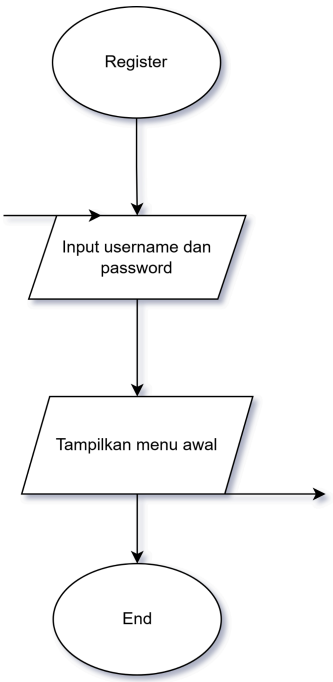
2025

1. Flowchart

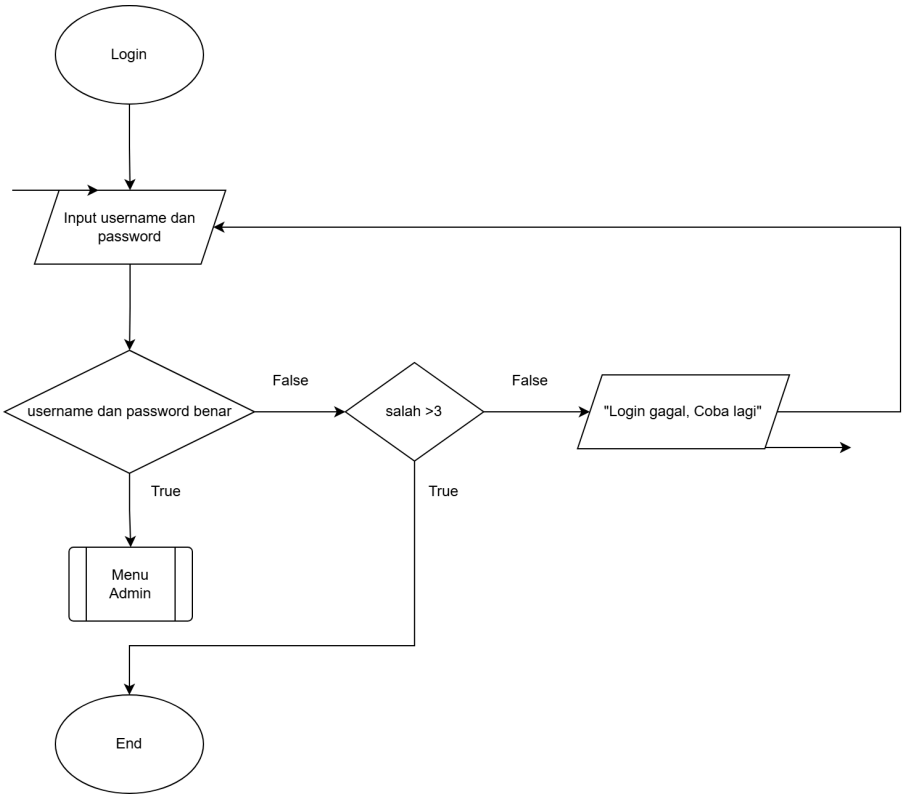
Flowchart ini membantu memvisualisasikan keputusan bercabang (pakai simbol belah ketupat) dan tindakan proses (pakai persegi panjang), sehingga mudah melihat jalur mana yang ditempuh sesuai kondisi.. Kemudian flowchart menggunakan predefined proses agar menunjukkan bahwa ada proses/fungsi yang terdefinisikan



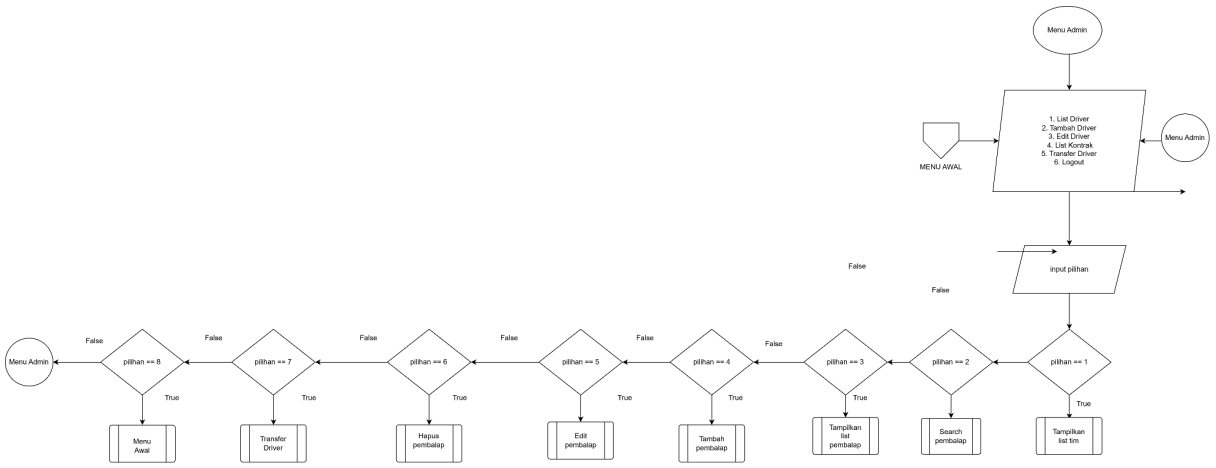
Gambar 1.1 Flowchart CRUD Menu awal



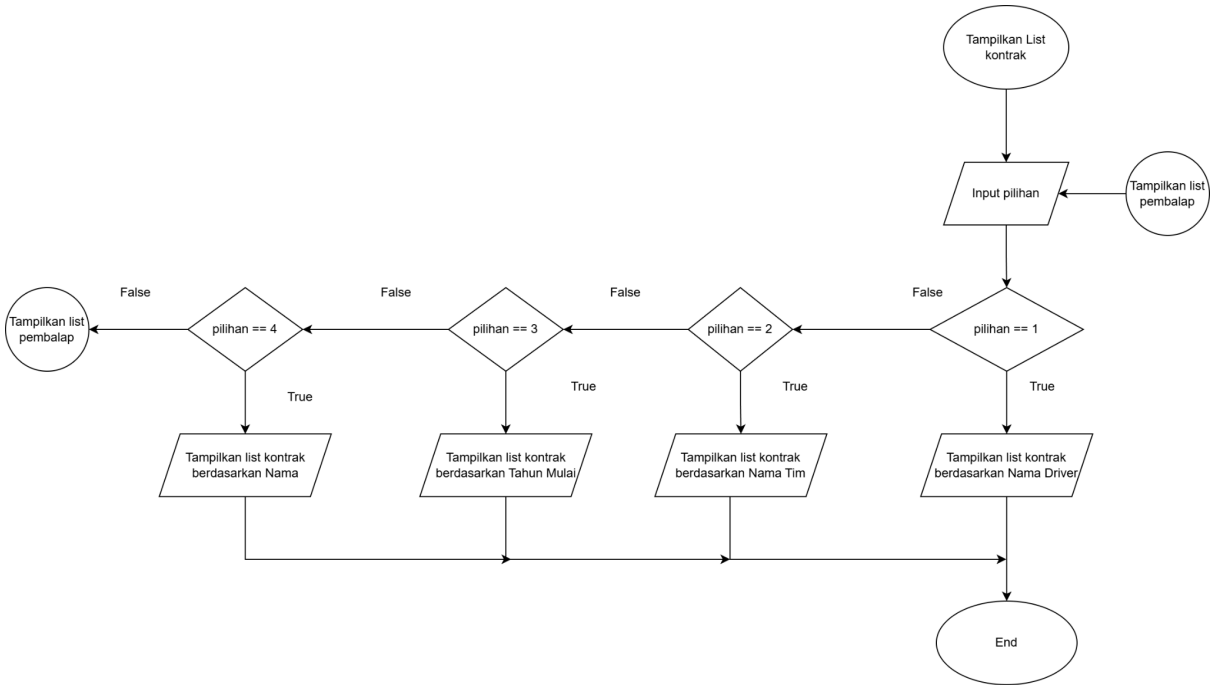
Gambar 1.2 Flowchart Register



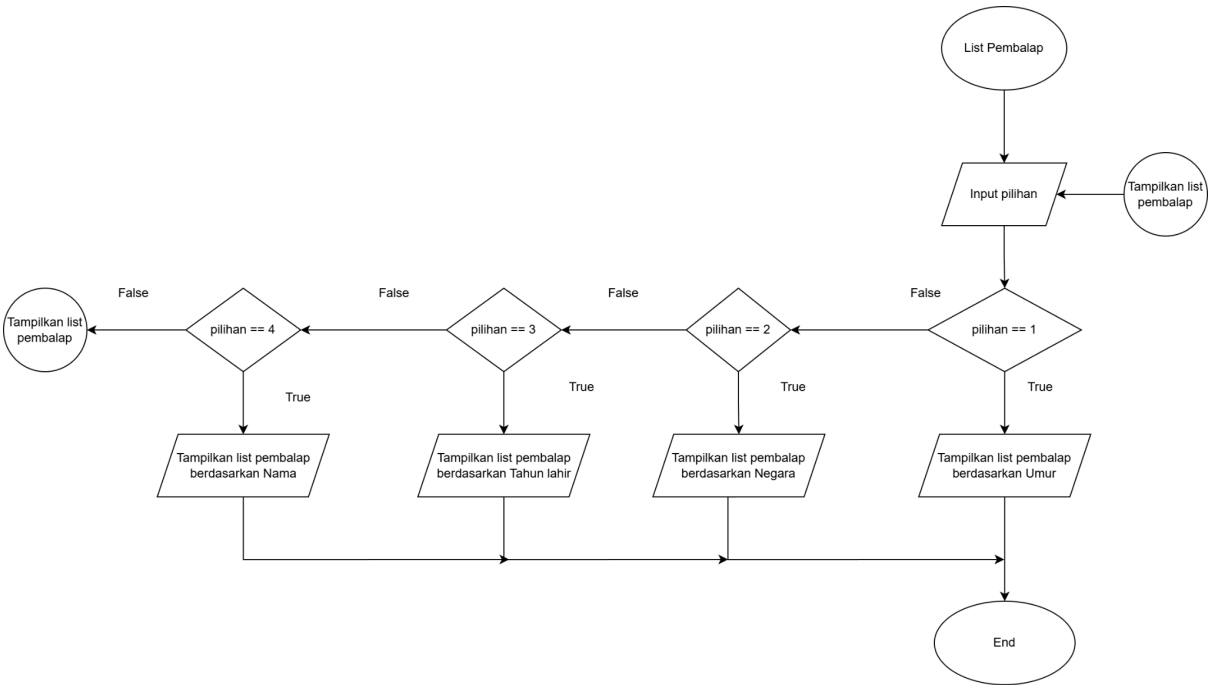
Gambar 1.3 Flowchart Login



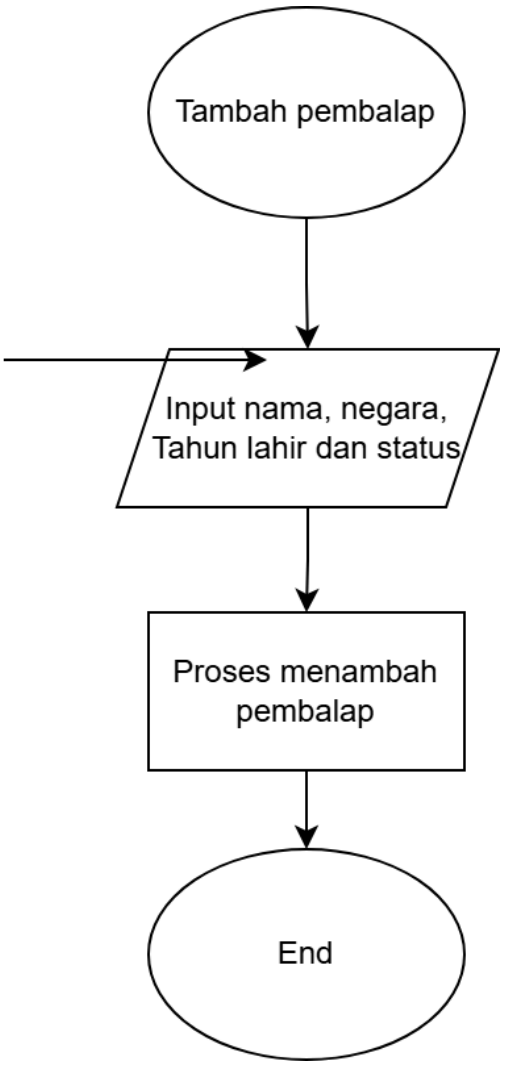
Gambar 1.4 Flowchart Menu Admin



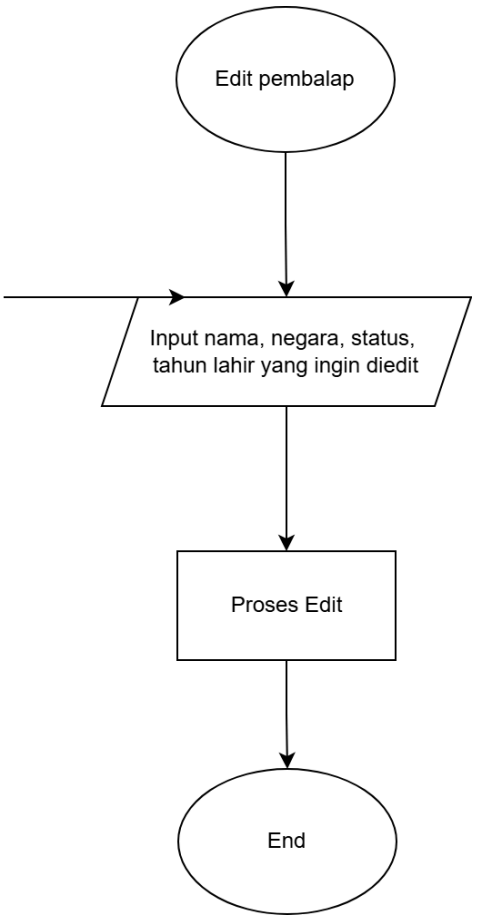
Gambar 1.5 Tampilkan List Kontrak



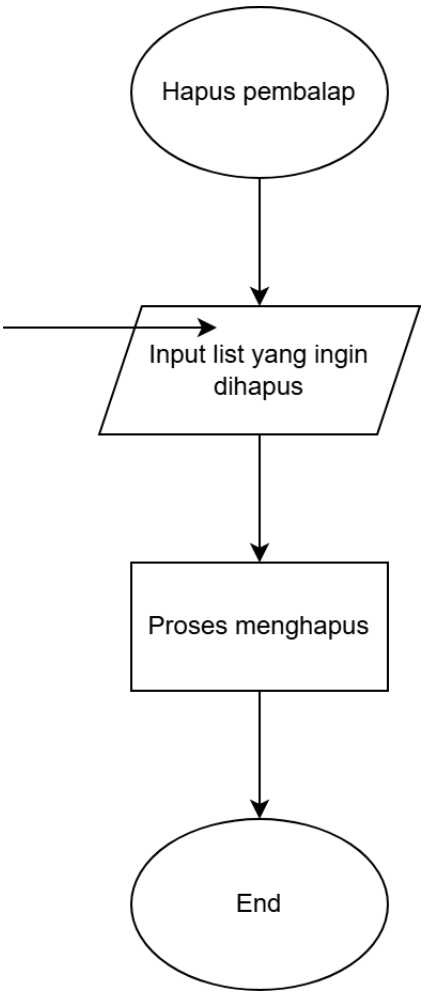
Gambar 1.6 Tampilkan list pembalap



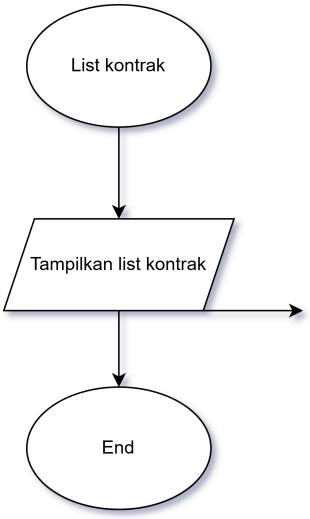
Gambar 1.7 Tambah pembalap



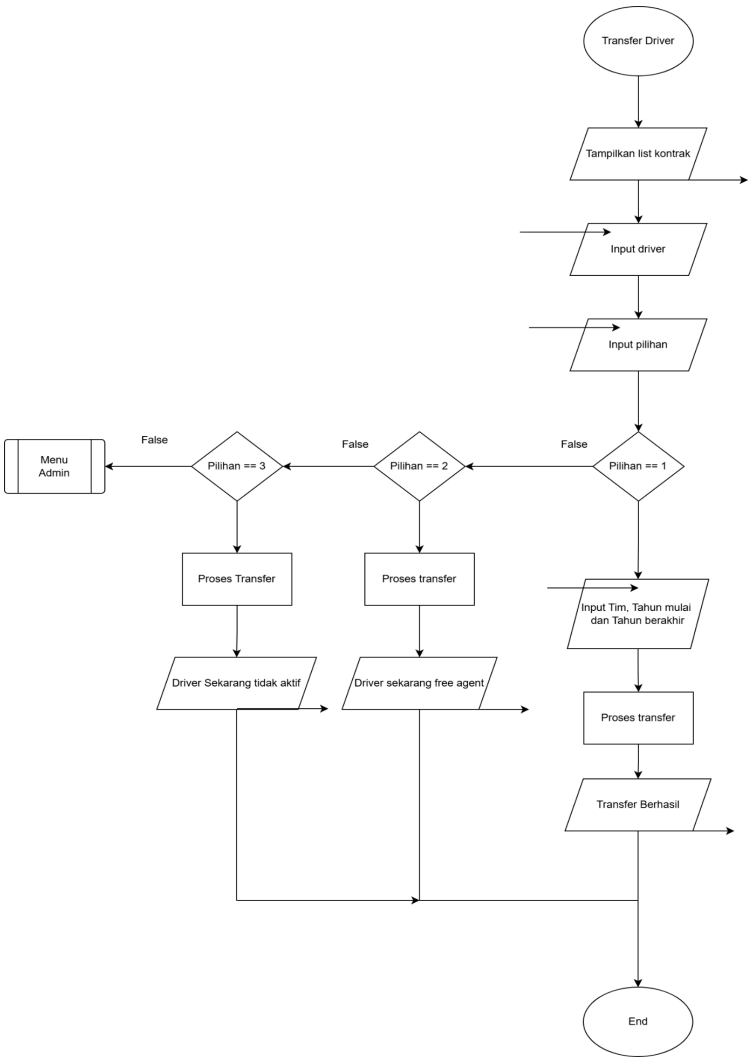
Gambar 1.8 Edit Pembalap



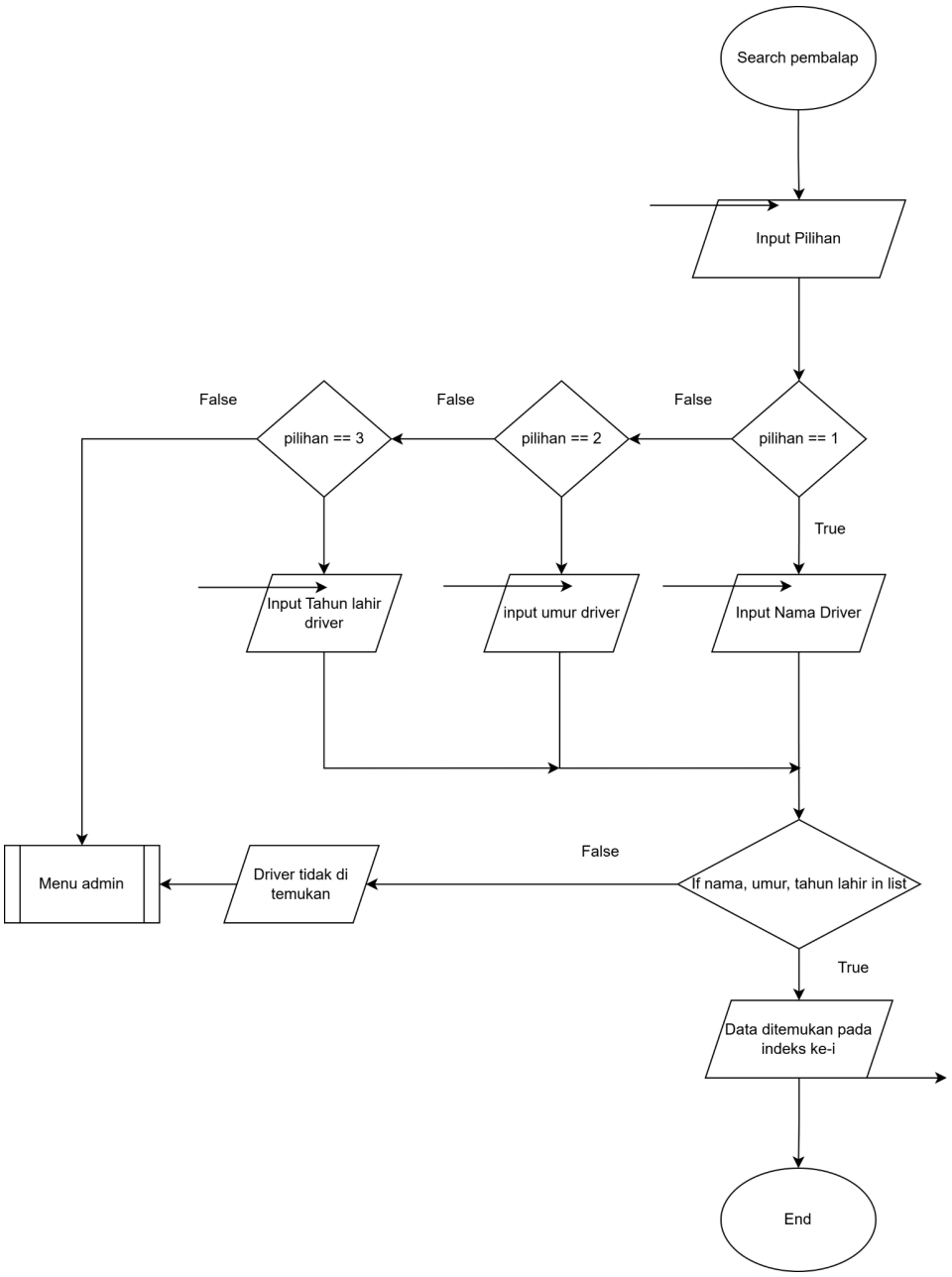
Gambar 1.9 Hapus pembalap



Gambar 1.10 List Kontrak



Gambar 1.11 Transfer Driver



Gambar 1.12 Menu Search Driver

2. Deskripsi Singkat Program

Secara keseluruhan, program ini berfungsi sebagai sistem manajemen kontrak dan informasi pembalap F1 yang sederhana, dengan kontrol akses berdasarkan role dan menu interaktif di konsol.

3. Source Code

```cpp
#include <iostream>
#include <string>
#include <iomanip>
#include <utility>
#include <limits>
#include <cctype>

using namespace std;

const int MAX_DRIVERS = 50;
const int MAX_USERS = 20;
const int MAX_TEAMS = 11;
const int MAX_CONTRACTS = 100;
const int CURRENT_YEAR = 2026;
const int RADIX_CHAR_RANGE = 27;

struct Driver {
    string name;
    string nationality;
    int birthYear;
    string status;
};

struct User {
    string username;
    string password;
};

struct Team {
    string name;
};

struct Contract {
    int driverIndex;
    int teamIndex;
    int startYear;
    int endYear;
};

// ================ HELPER ================

void clearInput() {
    cin.clear();
```

```
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
}

bool isValidDriverIndex(int index, int totalDrivers) {
    return index >= 0 && index < totalDrivers;
}

bool isValidTeamIndex(int index, int totalTeams) {
    return index >= 0 && index < totalTeams;
}

bool isContractActiveInYear(const Contract &contract, int year) {
    return contract.startYear <= year && contract.endYear >= year;
}

int getAge(const Driver &driver) {
    return CURRENT_YEAR - driver.birthYear;
}

string toUpperCase(string text) {
    for (string::size_type i = 0; i < text.length(); i++) {
        text[i] = static_cast<char>(toupper(static_cast<unsigned
char>(text[i])));
    }
    return text;
}

int getRadixCharValueFromLeft(const string &text, int posFromLeft) {
    int length = static_cast<int>(text.length());

    if (posFromLeft < 0 || posFromLeft >= length) {
        return 0;
    }

    unsigned char c = static_cast<unsigned char>(
        text[static_cast<string::size_type>(posFromLeft)]
    );

    c = static_cast<unsigned char>(toupper(c));

    if (c >= 'A' && c <= 'Z') {
        return static_cast<int>(c - 'A' + 1);
    }

    return 0;
}

int getMaxNameLengthDrivers(Driver drivers[], int totalDrivers) {
```

```
    int maxLen = 0;
    for (int i = 0; i < totalDrivers; i++) {
        int len = static_cast<int>(drivers[i].name.length());
        if (len > maxLen) {
            maxLen = len;
        }
    }
    return maxLen;
}

int getMaxNameLengthContractsByDriver(Contract contracts[], int
totalContracts, Driver drivers[]) {
    int maxLen = 0;
    for (int i = 0; i < totalContracts; i++) {
        int len =
static_cast<int>(drivers[contracts[i].driverIndex].name.length());
        if (len > maxLen) {
            maxLen = len;
        }
    }
    return maxLen;
}

int getMaxNameLengthContractsByTeam(Contract contracts[], int
totalContracts, Team teams[]) {
    int maxLen = 0;
    for (int i = 0; i < totalContracts; i++) {
        int len =
static_cast<int>(teams[contracts[i].teamIndex].name.length());
        if (len > maxLen) {
            maxLen = len;
        }
    }
    return maxLen;
}

// ================= QUICK SORT (ASCENDING) =================

int partitionNationalityAsc(Driver drivers[], int low, int high) {
    string pivot = drivers[high].nationality;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (drivers[j].nationality < pivot) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }
```

```
    swap(drivers[i + 1], drivers[high]);
    return i + 1;
}

void quickSortByNationalityAsc(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionNationalityAsc(drivers, low, high);
        quickSortByNationalityAsc(drivers, low, pivot - 1);
        quickSortByNationalityAsc(drivers, pivot + 1, high);
    }
}

int partitionBirthYearAsc(Driver drivers[], int low, int high) {
    int pivot = drivers[high].birthYear;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (drivers[j].birthYear < pivot) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }

    swap(drivers[i + 1], drivers[high]);
    return i + 1;
}

void quickSortByBirthYearAsc(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionBirthYearAsc(drivers, low, high);
        quickSortByBirthYearAsc(drivers, low, pivot - 1);
        quickSortByBirthYearAsc(drivers, pivot + 1, high);
    }
}

int partitionAgeAsc(Driver drivers[], int low, int high) {
    int pivot = getAge(drivers[high]);
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (getAge(drivers[j]) < pivot) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }

    swap(drivers[i + 1], drivers[high]);
```

```
    return i + 1;
}

void quickSortByAgeAsc(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionAgeAsc(drivers, low, high);
        quickSortByAgeAsc(drivers, low, pivot - 1);
        quickSortByAgeAsc(drivers, pivot + 1, high);
    }
}

int partitionContractStartYearAsc(Contract contracts[], int low, int high) {
    int pivot = contracts[high].startYear;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (contracts[j].startYear < pivot) {
            i++;
            swap(contracts[i], contracts[j]);
        }
    }

    swap(contracts[i + 1], contracts[high]);
    return i + 1;
}

void quickSortContractByStartYearAsc(Contract contracts[], int low, int
high) {
    if (low < high) {
        int pivot = partitionContractStartYearAsc(contracts, low, high);
        quickSortContractByStartYearAsc(contracts, low, pivot - 1);
        quickSortContractByStartYearAsc(contracts, pivot + 1, high);
    }
}

int partitionContractEndYearAsc(Contract contracts[], int low, int high) {
    int pivot = contracts[high].endYear;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (contracts[j].endYear < pivot) {
            i++;
            swap(contracts[i], contracts[j]);
        }
    }

    swap(contracts[i + 1], contracts[high]);
    return i + 1;
```

```
}

void quickSortContractByEndYearAsc(Contract contracts[], int low, int high)
{
    if (low < high) {
        int pivot = partitionContractEndYearAsc(contracts, low, high);
        quickSortContractByEndYearAsc(contracts, low, pivot - 1);
        quickSortContractByEndYearAsc(contracts, pivot + 1, high);
    }
}

// ================ BUBBLE SORT (DESCENDING) ================

void bubbleSortByAgeDesc(Driver drivers[], int totalDrivers) {
    for (int i = 0; i < totalDrivers - 1; i++) {
        for (int j = 0; j < totalDrivers - i - 1; j++) {
            if (getAge(drivers[j]) < getAge(drivers[j + 1])) {
                swap(drivers[j], drivers[j + 1]);
            }
        }
    }
}

void bubbleSortByBirthYearDesc(Driver drivers[], int totalDrivers) {
    for (int i = 0; i < totalDrivers - 1; i++) {
        for (int j = 0; j < totalDrivers - i - 1; j++) {
            if (drivers[j].birthYear < drivers[j + 1].birthYear) {
                swap(drivers[j], drivers[j + 1]);
            }
        }
    }
}

void bubbleSortByContractStartYearDesc(Contract contracts[], int
totalContracts) {
    for (int i = 0; i < totalContracts - 1; i++) {
        for (int j = 0; j < totalContracts - i - 1; j++) {
            if (contracts[j].startYear < contracts[j + 1].startYear) {
                swap(contracts[j], contracts[j + 1]);
            }
        }
    }
}

void bubbleSortByContractEndYearDesc(Contract contracts[], int
totalContracts) {
    for (int i = 0; i < totalContracts - 1; i++) {
        for (int j = 0; j < totalContracts - i - 1; j++) {
```

```
            if (contracts[j].endYear < contracts[j + 1].endYear) {
                swap(contracts[j], contracts[j + 1]);
            }
        }
    }
}

// ================= RADIX SORT NAMA =================

void countingSortDriversByName(Driver drivers[], int totalDrivers, int
posFromLeft) {
    Driver output[MAX_DRIVERS];
    int count[RADIX_CHAR_RANGE] = {0};

    for (int i = 0; i < totalDrivers; i++) {
        int key = getRadixCharValueFromLeft(drivers[i].name, posFromLeft);
        count[key]++;
    }

    for (int i = 1; i < RADIX_CHAR_RANGE; i++) {
        count[i] += count[i - 1];
    }

    for (int i = totalDrivers - 1; i >= 0; i--) {
        int key = getRadixCharValueFromLeft(drivers[i].name, posFromLeft);
        output[count[key] - 1] = drivers[i];
        count[key]--;
    }

    for (int i = 0; i < totalDrivers; i++) {
        drivers[i] = output[i];
    }
}

void radixSortDriversByName(Driver drivers[], int totalDrivers) {
    int maxLen = getMaxNameLengthDrivers(drivers, totalDrivers);

    for (int pos = maxLen - 1; pos >= 0; pos--) {
        countingSortDriversByName(drivers, totalDrivers, pos);
    }
}

void countingSortContractsByDriverName(Contract contracts[], int
totalContracts, Driver drivers[], int posFromLeft) {
    Contract output[MAX_CONTRACTS];
    int count[RADIX_CHAR_RANGE] = {0};

    for (int i = 0; i < totalContracts; i++) {
```

```
        int key =
getRadixCharValueFromLeft(drivers[contracts[i].driverIndex].name,
posFromLeft);
        count[key]++;
    }

    for (int i = 1; i < RADIX_CHAR_RANGE; i++) {
        count[i] += count[i - 1];
    }

    for (int i = totalContracts - 1; i >= 0; i--) {
        int key =
getRadixCharValueFromLeft(drivers[contracts[i].driverIndex].name,
posFromLeft);
        output[count[key] - 1] = contracts[i];
        count[key]--;
    }

    for (int i = 0; i < totalContracts; i++) {
        contracts[i] = output[i];
    }
}

void radixSortContractsByDriverName(Contract contracts[], int
totalContracts, Driver drivers[]) {
    int maxLen = getMaxNameLengthContractsByDriver(contracts,
totalContracts, drivers);

    for (int pos = maxLen - 1; pos >= 0; pos--) {
        countingSortContractsByDriverName(contracts, totalContracts,
drivers, pos);
    }
}

void countingSortContractsByTeamName(Contract contracts[], int
totalContracts, Team teams[], int posFromLeft) {
    Contract output[MAX_CONTRACTS];
    int count[RADIX_CHAR_RANGE] = {0};

    for (int i = 0; i < totalContracts; i++) {
        int key =
getRadixCharValueFromLeft(teams[contracts[i].teamIndex].name, posFromLeft);
        count[key]++;
    }

    for (int i = 1; i < RADIX_CHAR_RANGE; i++) {
        count[i] += count[i - 1];
    }
```

```
    for (int i = totalContracts - 1; i >= 0; i--) {
        int key =
getRadixCharValueFromLeft(teams[contracts[i].teamIndex].name, posFromLeft);
        output[count[key] - 1] = contracts[i];
        count[key]--;
    }

    for (int i = 0; i < totalContracts; i++) {
        contracts[i] = output[i];
    }
}

void radixSortContractsByTeamName(Contract contracts[], int totalContracts,
Team teams[]) {
    int maxLen = getMaxNameLengthContractsByTeam(contracts, totalContracts,
teams);

    for (int pos = maxLen - 1; pos >= 0; pos--) {
        countingSortContractsByTeamName(contracts, totalContracts, teams,
pos);
    }
}

// ================= INIT =================

void initializeDrivers(Driver drivers[], int &totalDrivers) {
    string names[] = {
        "Charles Leclerc", "Lewis Hamilton", "George Russell", "Kimi
Antonelli",
        "Lando Norris", "Oscar Piastri", "Fernando Alonso", "Lance Stroll",
        "Esteban Ocon", "Oliver Bearman", "Max Verstappen", "Sergio Perez",
        "Arvid Lindblad", "Pierre Gasly", "Valtteri Bottas", "Isack Hadjar",
        "Nico Hulkenberg", "Gabriel Bortoletto", "Liam Lawson", "Alexander
Albon",
        "Carlos Sainz", "Franco Colapinto"
    };

    string countries[] = {
        "Monaco", "UK", "UK", "Italy",
        "UK", "Australia", "Spain", "Canada",
        "France", "UK", "Netherlands", "Mexico",
        "Sweden", "France", "Finland", "France",
        "Germany", "Brazil", "New Zealand", "Thailand",
        "Spain", "Argentina"
    };

    int birthYears[] = {
```

```
        1997, 1985, 1998, 2006,
        1999, 2001, 1981, 1998,
        1996, 2005, 1997, 1990,
        2004, 1996, 1989, 2003,
        1987, 2004, 2002, 1996,
        1994, 2003
    };

    for (int i = 0; i < 22; i++) {
        drivers[i].name = names[i];
        drivers[i].nationality = countries[i];
        drivers[i].birthYear = birthYears[i];
        drivers[i].status = "Aktif";
    }

    totalDrivers = 22;
}

void initializeTeams(Team teams[], int &totalTeams) {
    string names[] = {
        "Ferrari", "Mercedes", "McLaren", "Red Bull", "RB",
        "Alpine", "Aston Martin", "Audi", "Williams", "Cadillac", "Haas"
    };

    for (int i = 0; i < 11; i++) {
        teams[i].name = names[i];
    }

    totalTeams = 11;
}

void initializeUsers(User users[], int &totalUsers) {
    users[0].username = "Attol";
    users[0].password = "2509106103";
    totalUsers = 1;
}

void initializeContracts(Contract contracts[], int &totalContracts) {
    int map[22] = {0, 0, 1, 1, 2, 2, 6, 6, 10, 10, 3, 9, 4, 5, 9, 3, 7, 7,
4, 8, 8, 5};

    for (int i = 0; i < 22; i++) {
        contracts[i].driverIndex = i;
        contracts[i].teamIndex = map[i];
        contracts[i].startYear = 2025;
        contracts[i].endYear = 2027;
    }
```

```
    totalContracts = 22;
}

// ================ USER ================

void registerUser(User users[], int &totalUsers) {
    if (totalUsers >= MAX_USERS) {
        cout << "Kapasitas user penuh!\n";
        return;
    }

    cout << "Username: ";
    getline(cin, users[totalUsers].username);

    cout << "Password: ";
    getline(cin, users[totalUsers].password);

    totalUsers++;
    cout << "Registrasi berhasil!\n";
}

bool loginUser(User users[], int totalUsers) {
    string username, password;

    cout << "Username: ";
    getline(cin, username);

    cout << "Password: ";
    getline(cin, password);

    for (int i = 0; i < totalUsers; i++) {
        if (users[i].username == username && users[i].password == password)
{
            return true;
        }
    }

    cout << "Login gagal!\n";
    return false;
}

// ================ DISPLAY ================

void displayDrivers(Driver drivers[], int totalDrivers) {
    cout << "\n=== LIST DRIVER ===\n";
    cout << left
         << setw(4)  << "No"
         << setw(25) << "Nama"
```

```
         << setw(15) << "Negara"
         << setw(12) << "Tahun Lahir"
         << setw(8)  << "Umur"
         << setw(15) << "Status" << '\n';

    cout << string(79, '=') << '\n';

    for (int i = 0; i < totalDrivers; i++) {
        cout << left
             << setw(4)  << i + 1
             << setw(25) << drivers[i].name
             << setw(15) << drivers[i].nationality
             << setw(12) << drivers[i].birthYear
             << setw(8)  << getAge(drivers[i])
             << setw(15) << drivers[i].status
             << '\n';
    }
}

void displayTeams(Team teams[], int totalTeams) {
    cout << "\n=== LIST TEAM ===\n";
    for (int i = 0; i < totalTeams; i++) {
        cout << i + 1 << ". " << teams[i].name << '\n';
    }
}

void displayAllContracts(Contract contracts[], int totalContracts, Driver
drivers[], Team teams[]) {
    cout << "\n=== LIST SEMUA KONTRAK ===\n";
    cout << left
         << setw(4)  << "No"
         << setw(25) << "Driver"
         << setw(20) << "Tim"
         << setw(12) << "Mulai"
         << setw(12) << "Selesai"
         << setw(15) << "Status 2026"
         << '\n';

    cout << string(88, '=') << '\n';

    for (int i = 0; i < totalContracts; i++) {
        string status2026 = isContractActiveInYear(contracts[i],
CURRENT_YEAR) ? "Aktif" : "Tidak Aktif";

        cout << left
             << setw(4)  << i + 1
             << setw(25) << drivers[contracts[i].driverIndex].name
             << setw(20) << teams[contracts[i].teamIndex].name
```

```
             << setw(12) << contracts[i].startYear
             << setw(12) << contracts[i].endYear
             << setw(15) << status2026
             << '\n';
    }

    if (totalContracts == 0) {
        cout << "Belum ada data kontrak.\n";
    }
}

void displayActiveContracts2026(Contract contracts[], int totalContracts,
Driver drivers[], Team teams[]) {
    cout << "\n=== LIST KONTRAK AKTIF TAHUN " << CURRENT_YEAR << " ===\n";
    cout << left
         << setw(4)  << "No"
         << setw(25) << "Driver"
         << setw(20) << "Tim"
         << setw(12) << "Mulai"
         << setw(12) << "Selesai"
         << '\n';

    cout << string(73, '=') << '\n';

    int nomor = 1;
    for (int i = 0; i < totalContracts; i++) {
        if (isContractActiveInYear(contracts[i], CURRENT_YEAR)) {
            cout << left
                 << setw(4)  << nomor
                 << setw(25) << drivers[contracts[i].driverIndex].name
                 << setw(20) << teams[contracts[i].teamIndex].name
                 << setw(12) << contracts[i].startYear
                 << setw(12) << contracts[i].endYear
                 << '\n';
            nomor++;
        }
    }

    if (nomor == 1) {
        cout << "Tidak ada kontrak aktif pada tahun " << CURRENT_YEAR <<
".\n";
    }
}

void displayFoundDriver(const Driver &driver, int index) {
    cout << "\nData ditemukan pada index ke-" << index << '\n';
    cout << left
         << setw(25) << "Nama" << ": " << driver.name << '\n'
```

```
         << setw(25) << "Negara" << ": " << driver.nationality << '\n'
         << setw(25) << "Tahun lahir" << ": " << driver.birthYear << '\n'
         << setw(25) << "Umur" << ": " << getAge(driver) << '\n'
         << setw(25) << "Status" << ": " << driver.status << '\n';
}

// ================ SEARCH ================

int jumpSearchByName(Driver *drivers, int totalDrivers, const string
&target) {
    if (totalDrivers <= 0) {
        return -1;
    }

    string targetUpper = toUpperCase(target);

    int step = 1;
    while (step * step < totalDrivers) {
        step++;
    }

    int prev = 0;
    int current = step;

    while (prev < totalDrivers &&
           toUpperCase((drivers + ((current < totalDrivers) ? current :
totalDrivers) - 1)->name) < targetUpper) {
        prev = current;
        current += step;

        if (prev >= totalDrivers) {
            return -1;
        }
    }

    int end = (current < totalDrivers) ? current : totalDrivers;

    for (int i = prev; i < end; i++) {
        if (toUpperCase((drivers + i)->name) == targetUpper) {
            return i;
        }
    }

    return -1;
}

int binarySearchByAge(Driver *drivers, int totalDrivers, int targetAge) {
    int left = 0;
```

```
    int right = totalDrivers - 1;

    while (left <= right) {
        int mid = left + (right - left) / 2;
        int ageMid = getAge(*(drivers + mid));

        if (ageMid == targetAge) {
            return mid;
        } else if (ageMid < targetAge) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }

    return -1;
}

void linearSearchByBirthYear(Driver *drivers, int totalDrivers, int
targetBirthYear) {
    bool found = false;

    cout << "\n=== HASIL PENCARIAN TAHUN LAHIR ===\n";
    for (int i = 0; i < totalDrivers; i++) {
        if ((drivers + i)->birthYear == targetBirthYear) {
            found = true;
            displayFoundDriver(*(drivers + i), i);
        }
    }

    if (!found) {
        cout << "Data dengan tahun lahir " << targetBirthYear << " tidak
ditemukan.\n";
    }
}

void searchDriverMenu(Driver drivers[], int totalDrivers) {
    int choice;

    cout << "\n=== SEARCH DRIVER ===\n";
    cout << "1. Nama Driver (Jump Search)\n";
    cout << "2. Umur (Binary Search)\n";
    cout << "3. Tahun Lahir (Linear Search)\n";
    cout << "Pilih: ";
    cin >> choice;

    if (cin.fail()) {
        clearInput();
```

```
        cout << "Input tidak valid!\n";
        return;
    }

    if (choice == 1) {
        clearInput();
        string targetName;
        cout << "Masukkan nama driver: ";
        getline(cin, targetName);

        Driver temp[MAX_DRIVERS];
        for (int i = 0; i < totalDrivers; i++) {
            temp[i] = drivers[i];
        }

        radixSortDriversByName(temp, totalDrivers);

        int index = jumpSearchByName(temp, totalDrivers, targetName);
        if (index != -1) {
            displayFoundDriver(temp[index], index);
        } else {
            cout << "Driver dengan nama \"" << targetName << "\" tidak
ditemukan.\n";
        }
    } else if (choice == 2) {
        int targetAge;
        cout << "Masukkan umur: ";
        cin >> targetAge;

        if (cin.fail()) {
            clearInput();
            cout << "Input umur tidak valid!\n";
            return;
        }

        Driver temp[MAX_DRIVERS];
        for (int i = 0; i < totalDrivers; i++) {
            temp[i] = drivers[i];
        }

        quickSortByAgeAsc(temp, 0, totalDrivers - 1);

        int index = binarySearchByAge(temp, totalDrivers, targetAge);
        if (index != -1) {
            displayFoundDriver(temp[index], index);
        } else {
            cout << "Driver dengan umur " << targetAge << " tidak
ditemukan.\n";
```

```
        }
    } else if (choice == 3) {
        int targetBirthYear;
        cout << "Masukkan tahun lahir: ";
        cin >> targetBirthYear;

        if (cin.fail()) {
            clearInput();
            cout << "Input tahun lahir tidak valid!\n";
            return;
        }

        linearSearchByBirthYear(drivers, totalDrivers, targetBirthYear);
    } else {
        cout << "Pilihan search tidak valid!\n";
    }
}

// ================= SORT MENU DRIVER =================

void sortDriversMenu(Driver drivers[], int totalDrivers) {
    int sortChoice;

    cout << "\n=== SORT DRIVER BY ===\n";
    cout << "1. Umur Descending (Bubble Sort)\n";
    cout << "2. Negara Ascending (Quick Sort)\n";
    cout << "3. Tahun Lahir Descending (Bubble Sort)\n";
    cout << "4. Nama Ascending (Radix Sort)\n";
    cout << "Pilih: ";
    cin >> sortChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    switch (sortChoice) {
        case 1:
            bubbleSortByAgeDesc(drivers, totalDrivers);
            break;
        case 2:
            quickSortByNationalityAsc(drivers, 0, totalDrivers - 1);
            break;
        case 3:
            bubbleSortByBirthYearDesc(drivers, totalDrivers);
            break;
        case 4:
```

```
            radixSortDriversByName(drivers, totalDrivers);
            break;
        default:
            cout << "Pilihan sort tidak valid!\n";
            return;
    }

    displayDrivers(drivers, totalDrivers);
}

// ================= SORT MENU KONTRAK =================

void sortContractsMenu(Contract contracts[], int totalContracts, Driver
drivers[], Team teams[]) {
    int sortChoice;
    int filterChoice;

    cout << "\n=== SORT KONTRAK BY ===\n";
    cout << "1. Nama Driver Ascending (Radix Sort)\n";
    cout << "2. Nama Tim Ascending (Radix Sort)\n";
    cout << "3. Tahun Mulai Descending (Bubble Sort)\n";
    cout << "4. Tahun Selesai Descending (Bubble Sort)\n";
    cout << "Pilih: ";
    cin >> sortChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    switch (sortChoice) {
        case 1:
            radixSortContractsByDriverName(contracts, totalContracts,
drivers);
            break;
        case 2:
            radixSortContractsByTeamName(contracts, totalContracts, teams);
            break;
        case 3:
            bubbleSortByContractStartYearDesc(contracts, totalContracts);
            break;
        case 4:
            bubbleSortByContractEndYearDesc(contracts, totalContracts);
            break;
        default:
            cout << "Pilihan sort tidak valid!\n";
            return;
```

```
    }

    cout << "\n=== TAMPILKAN KONTRAK ===\n";
    cout << "1. Semua kontrak\n";
    cout << "2. Hanya kontrak aktif tahun 2026\n";
    cout << "Pilih: ";
    cin >> filterChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    switch (filterChoice) {
        case 1:
            displayAllContracts(contracts, totalContracts, drivers, teams);
            break;
        case 2:
            displayActiveContracts2026(contracts, totalContracts, drivers,
teams);
            break;
        default:
            cout << "Pilihan tampilan tidak valid!\n";
            break;
    }
}

// ================ CRUD DRIVER ================

void addDriver(Driver drivers[], int &totalDrivers) {
    if (totalDrivers >= MAX_DRIVERS) {
        cout << "Data driver penuh!\n";
        return;
    }

    clearInput();

    cout << "Nama: ";
    getline(cin, drivers[totalDrivers].name);

    cout << "Negara: ";
    getline(cin, drivers[totalDrivers].nationality);

    cout << "Tahun lahir: ";
    cin >> drivers[totalDrivers].birthYear;

    if (cin.fail()) {
```

```
        clearInput();
        cout << "Input tahun lahir tidak valid!\n";
        return;
    }

    clearInput();

    cout << "Status: ";
    getline(cin, drivers[totalDrivers].status);

    totalDrivers++;
    cout << "Driver berhasil ditambahkan!\n";
}

void editDriver(Driver drivers[], int totalDrivers) {
    if (totalDrivers == 0) {
        cout << "Tidak ada data driver.\n";
        return;
    }

    displayDrivers(drivers, totalDrivers);

    int choice;
    cout << "Pilih nomor driver yang ingin diedit: ";
    cin >> choice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    int index = choice - 1;
    if (!isValidDriverIndex(index, totalDrivers)) {
        cout << "Nomor driver tidak valid!\n";
        return;
    }

    clearInput();

    cout << "Nama baru: ";
    getline(cin, drivers[index].name);

    cout << "Negara baru: ";
    getline(cin, drivers[index].nationality);

    cout << "Tahun lahir baru: ";
    cin >> drivers[index].birthYear;
```

```
    if (cin.fail()) {
        clearInput();
        cout << "Input tahun lahir tidak valid!\n";
        return;
    }

    clearInput();

    cout << "Status baru: ";
    getline(cin, drivers[index].status);

    cout << "Data driver berhasil diupdate!\n";
}

void deleteDriverContracts(Contract contracts[], int &totalContracts, int
driverIndex) {
    for (int i = 0; i < totalContracts; i++) {
        if (contracts[i].driverIndex == driverIndex) {
            for (int j = i; j < totalContracts - 1; j++) {
                contracts[j] = contracts[j + 1];
            }
            totalContracts--;
            i--;
        }
    }
}

void adjustContractDriverIndexes(Contract contracts[], int totalContracts,
int deletedDriverIndex) {
    for (int i = 0; i < totalContracts; i++) {
        if (contracts[i].driverIndex > deletedDriverIndex) {
            contracts[i].driverIndex--;
        }
    }
}

void deleteDriver(Driver drivers[], int &totalDrivers, Contract contracts[],
int &totalContracts) {
    if (totalDrivers == 0) {
        cout << "Tidak ada data driver.\n";
        return;
    }

    displayDrivers(drivers, totalDrivers);

    int choice;
    cout << "Pilih nomor driver yang ingin dihapus: ";
```

```
    cin >> choice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    int index = choice - 1;
    if (!isValidDriverIndex(index, totalDrivers)) {
        cout << "Nomor driver tidak valid!\n";
        return;
    }

    deleteDriverContracts(contracts, totalContracts, index);
    adjustContractDriverIndexes(contracts, totalContracts, index);

    for (int i = index; i < totalDrivers - 1; i++) {
        drivers[i] = drivers[i + 1];
    }

    totalDrivers--;
    cout << "Driver berhasil dihapus!\n";
}

// ================ KONTRAK ================

void addContract(Driver drivers[], int totalDrivers,
                 Team teams[], int totalTeams,
                 Contract contracts[], int &totalContracts) {
    if (totalContracts >= MAX_CONTRACTS) {
        cout << "Data kontrak penuh!\n";
        return;
    }

    int driverChoice;
    int teamChoice;
    int startYear;
    int endYear;

    displayDrivers(drivers, totalDrivers);
    cout << "Pilih driver: ";
    cin >> driverChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input driver tidak valid!\n";
        return;
```

```
    }

    int driverIndex = driverChoice - 1;
    if (!isValidDriverIndex(driverIndex, totalDrivers)) {
        cout << "Pilihan driver tidak valid!\n";
        return;
    }

    displayTeams(teams, totalTeams);
    cout << "Pilih tim: ";
    cin >> teamChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tim tidak valid!\n";
        return;
    }

    int teamIndex = teamChoice - 1;
    if (!isValidTeamIndex(teamIndex, totalTeams)) {
        cout << "Pilihan tim tidak valid!\n";
        return;
    }

    cout << "Tahun mulai: ";
    cin >> startYear;
    if (cin.fail()) {
        clearInput();
        cout << "Input tahun mulai tidak valid!\n";
        return;
    }

    cout << "Tahun selesai: ";
    cin >> endYear;
    if (cin.fail()) {
        clearInput();
        cout << "Input tahun selesai tidak valid!\n";
        return;
    }

    if (endYear < startYear) {
        cout << "Tahun selesai tidak boleh lebih kecil dari tahun mulai!\n";
        return;
    }

    contracts[totalContracts].driverIndex = driverIndex;
    contracts[totalContracts].teamIndex = teamIndex;
    contracts[totalContracts].startYear = startYear;
```

```
    contracts[totalContracts].endYear = endYear;
    totalContracts++;

    drivers[driverIndex].status = "Aktif";

    cout << "Kontrak berhasil ditambahkan!\n";
}

void transferDriver(Driver drivers[], int totalDrivers,
                    Team teams[], int totalTeams,
                    Contract contracts[], int &totalContracts) {
    addContract(drivers, totalDrivers, teams, totalTeams, contracts,
totalContracts);
}

// ================ MENU ADMIN ================

void adminMenu(Driver drivers[], int &totalDrivers,
               Team teams[], int totalTeams,
               Contract contracts[], int &totalContracts) {
    int choice;

    do {
        cout << "\n=== MENU ADMIN ===\n";
        cout << "1. List Driver\n";
        cout << "2. Search Driver\n";
        cout << "3. Tambah Driver\n";
        cout << "4. Edit Driver\n";
        cout << "5. Hapus Driver\n";
        cout << "6. List Kontrak\n";
        cout << "7. Tambah / Transfer Kontrak\n";
        cout << "8. Logout\n";
        cout << "Pilih: ";
        cin >> choice;

        if (cin.fail()) {
            clearInput();
            cout << "Input tidak valid!\n";
            continue;
        }

        switch (choice) {
            case 1:
                sortDriversMenu(drivers, totalDrivers);
                break;
            case 2:
                searchDriverMenu(drivers, totalDrivers);
                break;
```

```
            case 3:
                addDriver(drivers, totalDrivers);
                break;
            case 4:
                editDriver(drivers, totalDrivers);
                break;
            case 5:
                deleteDriver(drivers, totalDrivers, contracts,
totalContracts);
                break;
            case 6:
                if (totalContracts > 0) {
                    sortContractsMenu(contracts, totalContracts, drivers,
teams);
                } else {
                    cout << "Tidak ada data kontrak.\n";
                }
                break;
            case 7:
                transferDriver(drivers, totalDrivers, teams, totalTeams,
contracts, totalContracts);
                break;
            case 8:
                cout << "Logout...\n";
                break;
            default:
                cout << "Menu belum tersedia.\n";
        }

    } while (choice != 8);
}

// ================ MAIN ================

int main() {
    Driver drivers[MAX_DRIVERS];
    User users[MAX_USERS];
    Team teams[MAX_TEAMS];
    Contract contracts[MAX_CONTRACTS];

    int totalDrivers = 0;
    int totalUsers = 0;
    int totalTeams = 0;
    int totalContracts = 0;

    initializeDrivers(drivers, totalDrivers);
    initializeTeams(teams, totalTeams);
    initializeUsers(users, totalUsers);
```

```
    initializeContracts(contracts, totalContracts);

    int choice;

    do {
        cout << "\n=== MENU AWAL ===\n";
        cout << "1. Register\n";
        cout << "2. Login\n";
        cout << "3. Keluar\n";
        cout << "Pilih: ";
        cin >> choice;

        if (cin.fail()) {
            clearInput();
            cout << "Input tidak valid!\n";
            continue;
        }

        clearInput();

        if (choice == 1) {
            registerUser(users, totalUsers);
        } else if (choice == 2) {
            if (loginUser(users, totalUsers)) {
                adminMenu(drivers, totalDrivers, teams, totalTeams,
contracts, totalContracts);
            }
        } else if (choice == 3) {
            cout << "Program selesai.\n";
        } else {
            cout << "Pilihan tidak valid.\n";
        }

    } while (choice != 3);

    return 0;
}
```

4. Hasil Output

```
+-------------------------------------------+
|      SELAMAT DATANG DI BURSA TRANSFER F1  |
+-------------------------------------------+

================================================
MENU AWAL
================================================
+-----------------------------------+
| 1.  Register Admin                |
| 2.  Login Admin                   |
| 3.  Keluar                        |
+-----------------------------------+
Pilihan : 2
Username : Attol
Password : 2509106103
Login berhasil.
```

Gambar 4.1 Hasil Output Menu Awal

```
=== MENU ===
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Transfer Driver
7. Logout
Pilih:
```

Gambar 4.2 Hasil Output Login

```
================================================================================
DATA PEMBALAP LENGKAP
================================================================================
+----+-------------------------+---------------+------------+-------+------------+
| No | Nama                    | Negara        | Tahun Lahir | Umur  | Status     |
+----+-------------------------+---------------+------------+-------+------------+
| 1  | Charles Leclerc         | Monaco        | 1997       | 29    | Aktif      |
| 2  | Lewis Hamilton          | UK            | 1985       | 41    | Aktif      |
| 3  | George Russell          | UK            | 1998       | 28    | Aktif      |
| 4  | Kimi Antonelli          | Italy         | 2006       | 20    | Aktif      |
| 5  | Lando Norris            | UK            | 1999       | 27    | Aktif      |
| 6  | Oscar Piastri           | Australia     | 2001       | 25    | Aktif      |
| 7  | Fernando Alonso         | Spain         | 1981       | 45    | Aktif      |
| 8  | Lance Stroll            | Canada        | 1998       | 28    | Aktif      |
| 9  | Esteban Ocon            | France        | 1996       | 30    | Aktif      |
| 10 | Oliver Bearman          | UK            | 2005       | 21    | Aktif      |
| 11 | Isack Hadjar            | France        | 2003       | 23    | Aktif      |
| 12 | Max Verstappen          | Netherlands   | 1997       | 29    | Aktif      |
| 13 | Carlos Sainz Jr.        | Spain         | 1994       | 32    | Aktif      |
| 14 | Alexander Albon         | UK            | 1996       | 30    | Aktif      |
| 15 | Valtteri Bottas         | Finland       | 1989       | 37    | Aktif      |
| 16 | Sergio Perez            | Mexico        | 1990       | 36    | Aktif      |
| 17 | Nico Hulkenberg         | Germany       | 1987       | 39    | Aktif      |
| 18 | Gabriel Bortoleto       | Brazil        | 2004       | 22    | Aktif      |
| 19 | Pierre Gasly            | France        | 1996       | 30    | Aktif      |
| 20 | Franco Colapinto        | Argentina     | 1999       | 27    | Aktif      |
| 21 | Liam Lawson             | New Zealand   | 2002       | 24    | Aktif      |
| 22 | Arvid Lindblad          | Sweden        | 2004       | 22    | Aktif      |
+----+-------------------------+---------------+------------+-------+------------+
```

Gambar 4.3 Hasil Output List pembalap

```
=== MENU ===
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Transfer Driver
7. Logout
Pilih: 2
Nama: Benjamin Seskowi
Negara: Solovenia
Tahun lahir: 2002
Status (Aktif/Tidak Aktif/Free Agent): Aktif
Driver berhasil ditambahkan!
```

Gambar 4.4 Output Create Pembalap

```
=== LIST DRIVER ===
No  Nama                 Negara        Tahun Lahir Umur    Status
=============================================================================
1   Fernando Alonso      Spain         1981        45      Aktif
2   Lewis Hamilton       UK            1985        41      Aktif
3   Nico Hulkenberg      Germany       1987        39      Aktif
4   Valtteri Bottas      Finland       1989        37      Aktif
5   Sergio Perez         Mexico        1990        36      Aktif
6   Carlos Sainz         Spain         1994        32      Aktif
7   Alexander Albon      Thailand      1996        30      Aktif
8   Esteban Ocon         France        1996        30      Aktif
9   Pierre Gasly         France        1996        30      Aktif
10  Charles Leclerc      Monaco        1997        29      Aktif
11  Max Verstappen       Netherlands   1997        29      Aktif
12  George Russell       UK            1998        28      Aktif
13  Lance Stroll         Canada        1998        28      Aktif
14  Lando Norris         UK            1999        27      Aktif
15  Oscar Piastri        Australia     2001        25      Aktif
16  Liam Lawson          New Zealand   2002        24      Aktif
17  Franco Colapinto     Argentina     2003        23      Aktif
18  Isack Hadjar         France        2003        23      Aktif
19  Arvid Lindblad       Sweden        2004        22      Aktif
20  Gabriel Bortoletto   Brazil        2004        22      Aktif
21  Oliver Bearman       UK            2005        21      Aktif
22  Kimi Antonelli       Italy         2006        20      Aktif
```

Gambar 4.5 CRUD list pembalap sort berdasarkan umur(Descending) (Bubble Sort)

```
=== LIST DRIVER ===
No  Nama                 Negara        Tahun Lahir Umur    Status
=============================================================================
1   Franco Colapinto     Argentina     2003        23      Aktif
2   Oscar Piastri        Australia     2001        25      Aktif
3   Gabriel Bortoletto   Brazil        2004        22      Aktif
4   Lance Stroll         Canada        1998        28      Aktif
5   Valtteri Bottas      Finland       1989        37      Aktif
6   Pierre Gasly         France        1996        30      Aktif
7   Esteban Ocon         France        1996        30      Aktif
8   Isack Hadjar         France        2003        23      Aktif
9   Nico Hulkenberg      Germany       1987        39      Aktif
10  Kimi Antonelli       Italy         2006        20      Aktif
11  Sergio Perez         Mexico        1990        36      Aktif
12  Charles Leclerc      Monaco        1997        29      Aktif
13  Max Verstappen       Netherlands   1997        29      Aktif
14  Liam Lawson          New Zealand   2002        24      Aktif
15  Fernando Alonso      Spain         1981        45      Aktif
16  Carlos Sainz         Spain         1994        32      Aktif
17  Arvid Lindblad       Sweden        2004        22      Aktif
18  Alexander Albon      Thailand      1996        30      Aktif
19  George Russell       UK            1998        28      Aktif
20  Lewis Hamilton       UK            1985        41      Aktif
21  Lando Norris         UK            1999        27      Aktif
22  Oliver Bearman       UK            2005        21      Aktif
```

Gambar 4.6 List sort berdasarkan Negara (Quick Sort)

```
=== LIST DRIVER ===
No  Nama                 Negara          Tahun Lahir Umur    Status
================================================================================
1   Kimi Antonelli       Italy           2006        20      Aktif
2   Oliver Bearman       UK              2005        21      Aktif
3   Arvid Lindblad       Sweden          2004        22      Aktif
4   Gabriel Bortoletto   Brazil          2004        22      Aktif
5   Isack Hadjar         France          2003        23      Aktif
6   Franco Colapinto     Argentina       2003        23      Aktif
7   Liam Lawson          New Zealand     2002        24      Aktif
8   Oscar Piastri        Australia       2001        25      Aktif
9   Lando Norris         UK              1999        27      Aktif
10  Lance Stroll         Canada          1998        28      Aktif
11  George Russell       UK              1998        28      Aktif
12  Max Verstappen       Netherlands     1997        29      Aktif
13  Charles Leclerc      Monaco          1997        29      Aktif
14  Alexander Albon      Thailand        1996        30      Aktif
15  Pierre Gasly         France          1996        30      Aktif
16  Esteban Ocon         France          1996        30      Aktif
17  Carlos Sainz         Spain           1994        32      Aktif
18  Sergio Perez         Mexico          1990        36      Aktif
19  Valtteri Bottas      Finland         1989        37      Aktif
20  Nico Hulkenberg      Germany         1987        39      Aktif
21  Lewis Hamilton       UK              1985        41      Aktif
22  Fernando Alonso      Spain           1981        45      Aktif
```

Gambar 4.7 Output sort berdasarkan tahun lahir (Bubble sort)

```
=== LIST DRIVER ===
No  Nama                 Negara        Tahun Lahir Umur    Status
================================================================================
1   Alexander Albon      Thailand      1996        30      Aktif
2   Arvid Lindblad       Sweden        2004        22      Aktif
3   Carlos Sainz         Spain         1994        32      Aktif
4   Charles Leclerc      Monaco        1997        29      Aktif
5   Esteban Ocon         France        1996        30      Aktif
6   Fernando Alonso      Spain         1981        45      Aktif
7   Franco Colapinto     Argentina     2003        23      Aktif
8   Gabriel Bortoletto   Brazil        2004        22      Aktif
9   George Russell       UK            1998        28      Aktif
10  Isack Hadjar         France        2003        23      Aktif
11  Kimi Antonelli       Italy         2006        20      Aktif
12  Lance Stroll         Canada        1998        28      Aktif
13  Lando Norris         UK            1999        27      Aktif
14  Lewis Hamilton       UK            1985        41      Aktif
15  Liam Lawson          New Zealand   2002        24      Aktif
16  Max Verstappen       Netherlands   1997        29      Aktif
17  Nico Hulkenberg      Germany       1987        39      Aktif
18  Oliver Bearman       UK            2005        21      Aktif
19  Oscar Piastri        Australia     2001        25      Aktif
20  Pierre Gasly         France        1996        30      Aktif
21  Sergio Perez         Mexico        1990        36      Aktif
22  Valtteri Bottas      Finland       1989        37      Aktif
```

4.8 List Sort berdasarkan Urutan nama (A-Z) (Radix Sort)

```
================================================================================
DATA PEMBALAP LENGKAP
================================================================================
+----+--------------------------+----------------+-------------+-------+-------------+
| No | Nama                     | Negara         | Tahun Lahir | Umur  | Status      |
+----+--------------------------+----------------+-------------+-------+-------------+
| 1  | Charles Leclerc          | Monaco         | 1997        | 29    | Aktif       |
| 2  | Lewis Hamilton           | UK             | 1985        | 41    | Aktif       |
| 3  | George Russell           | UK             | 1998        | 28    | Aktif       |
| 4  | Kimi Antonelli           | Italy          | 2006        | 20    | Aktif       |
| 5  | Lando Norris             | UK             | 1999        | 27    | Aktif       |
| 6  | Oscar Piastri            | Australia      | 2001        | 25    | Aktif       |
| 7  | Fernando Alonso          | Spain          | 1981        | 45    | Aktif       |
| 8  | Lance Stroll             | Canada         | 1998        | 28    | Aktif       |
| 9  | Esteban Ocon             | France         | 1996        | 30    | Aktif       |
| 10 | Oliver Bearman           | UK             | 2005        | 21    | Aktif       |
| 11 | Isack Hadjar             | France         | 2003        | 23    | Aktif       |
| 12 | Max Verstappen           | Netherlands    | 1997        | 29    | Aktif       |
| 13 | Carlos Sainz Jr.         | Spain          | 1994        | 32    | Aktif       |
| 14 | Alexander Albon          | UK             | 1996        | 30    | Aktif       |
| 15 | Valtteri Bottas          | Finland        | 1989        | 37    | Aktif       |
| 16 | Sergio Perez             | Mexico         | 1990        | 36    | Aktif       |
| 17 | Nico Hulkenberg          | Germany        | 1987        | 39    | Aktif       |
| 18 | Gabriel Bortoleto        | Brazil         | 2004        | 22    | Aktif       |
| 19 | Pierre Gasly             | France         | 1996        | 30    | Aktif       |
| 20 | Franco Colapinto         | Argentina      | 1999        | 27    | Aktif       |
| 21 | Liam Lawson              | New Zealand    | 2002        | 24    | Aktif       |
| 22 | Arvid Lindblad           | Sweden         | 2004        | 22    | Aktif       |
| 23 | Benjamin Seskowi         | Solovenia      | 2002        | 24    | Tidak Aktif |
+----+--------------------------+----------------+-------------+-------+-------------+
Pilih nomor pembalap yang ingin dihapus : 23
Pembalap berhasil dihapus.
```

4.9 Output Delete Pembalap

```
| 22 | Arvid Lindblad           | Sweden         | 2004        | 22    | Aktif       |
+----+--------------------------+----------------+-------------+-------+-------------+
Pilih nomor pembalap yang ingin diedit : 22
Nama baru        : Arvid Limbad
Negara baru      : Swedia
Tahun lahir baru : 2007
Pilih status baru:
1. Aktif
2. Tidak Aktif
Pilihan          : 1
Pembalap berhasil diedit.
```

Gambar 4.10 Edit Pembalap

```
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Tambah / Transfer Kontrak
7. Logout
Pilih: 5

=== SORT KONTRAK BY ===
1. Nama Driver (A-Z)
2. Nama Tim (A-Z)
3. Tahun Mulai (Descending)
4. Tahun Selesai (Descending)
Pilih: 1

=== TAMPILKAN KONTRAK ===
1. Semua kontrak
2. Hanya kontrak aktif tahun 2026
Pilih: 1

=== LIST SEMUA KONTRAK ===
No  Driver                Tim                Mulai      Selesai    Status 2026
================================================================================
1   Alexander Albon       Ferrari            2025       2027       Aktif
2   Arvid Lindblad        Ferrari            2025       2027       Aktif
3   Carlos Sainz          Mercedes           2025       2027       Aktif
4   Charles Leclerc       Mercedes           2025       2027       Aktif
5   Esteban Ocon          McLaren            2025       2027       Aktif
6   Fernando Alonso       McLaren            2025       2027       Aktif
7   Franco Colapinto      Aston Martin       2025       2027       Aktif
8   Gabriel Bortoletto    Aston Martin       2025       2027       Aktif
```

Gambar 4.11 Output List Kontrak

```
=== SEARCH DRIVER ===
1. Nama Driver (Fibonacci Search)
2. Umur (Binary Search)
3. Tahun Lahir (Linear Search)
Pilih: 3
Masukkan tahun lahir: 2004

=== HASIL PENCARIAN TAHUN LAHIR ===

Data ditemukan pada index ke-12
Nama                    : Arvid Lindblad
Negara                  : Sweden
Tahun lahir             : 2004
Umur                    : 22
Status                  : Aktif

Data ditemukan pada index ke-17
Nama                    : Gabriel Bortoletto
Negara                  : Brazil
Tahun lahir             : 2004
Umur                    : 22
Status                  : Aktif
```

Gambar 4.12 Output Search Berdasarkan Tahun Lahir (Linier Search)

```
=== SEARCH DRIVER ===
1. Nama Driver (Fibonacci Search)
2. Umur (Binary Search)
3. Tahun Lahir (Linear Search)
Pilih: 2
Masukkan umur: 30

Data ditemukan pada index ke-13
Nama                    : Esteban Ocon
Negara                  : France
Tahun lahir             : 1996
Umur                    : 30
Status                  : Aktif
```

Gambar 4.13 Output search berdasarkan umur (Binary Search)

```
=== SEARCH DRIVER ===
1. Nama Driver (Fibonacci Search)
2. Umur (Binary Search)
3. Tahun Lahir (Linear Search)
Pilih: 1
Masukkan nama driver: Charles Leclerc

Data ditemukan pada index ke-3
Nama                    : Charles Leclerc
Negara                  : Monaco
Tahun lahir             : 1997
Umur                    : 29
Status                  : Aktif
```

Gambar 4.14 Output search berdasarkan Nama (Jump Search)

5. Langkah-langkah GIT

```
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git add .
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git commit -m "KOMAT KAMIT"
[main aede32e] KOMAT KAMIT
 3 files changed, 233 insertions(+)
 create mode 100644 post-test/post-test-7/2509106103-KhayzanDwitraAttala-PT-7 (1).pdf
 create mode 100644 post-test/post-test-7/2509106103-PT-7.drawio.png
 create mode 100644 post-test/post-test-7/2509106103_KhayzanDwitraAttala-PT-7.py
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 868.02 KiB | 27.13 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Khayzannn/praktikum-apd.git
   609b85f..aede32e  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7>
```

Gambar 5.1 Langkah - Langkah Git

5.1 GIT Add

Git add adalah perintah diGit yang digunakan untuk menambahkan perubahan pada berkas di direktori kerja ke area staging (atau indeks), mempersiapkannya untuk dimasukkan ke dalam commit berikutnya.

5.2 GIT Commit

Git commit adalah perintah di sistem kontrol versi Git untuk menyimpan (melakukan "commit") serangkaian perubahan yang telah dipilih (di-staging) ke dalam repositori lokal Anda, membuat sebuah "snapshot" atau titik pemeriksaan pada riwayat proyek dengan deskripsi perubahan.

5.3 GIT Push

Git push adalah perintah dalam Git yang digunakan untuk mengunggah (mengirimkan) perubahan yang telah dicatat di repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga memperbarui cabang tersebut dan membagikan pekerjaan dengan tim atau menyinkronkan dengan versi pusat.